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Leveraging Machine Learning to Identify Effective Teaching Strategies for Children with Autism Spectrum Disorder

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***Abstract*—Autism Spectrum Disorder (ASD) affects children globally, with rising prevalence underscoring the need for early and effective educational interventions. This study evaluates the use of machine learning (ML) algorithms to identify personalized teaching methods for children with ASD. We consolidated multiple datasets into a single, comprehensive dataset comprising 3043 samples with 17 features, which consists of behavioral and individual characteristics, further categorized into seven different teaching methods. We have used four ML algorithms: K-Nearest Neighbors (KNN), Multi-Layer Perceptron (MLP), Random Forest, and Decision Tree, to predict the most suitable educational strategies. Our results indicate that MLP and Decision Tree algorithms exhibited the highest accuracy in recommending tailored teaching methods. This study underscores the potential of machine learning in tailoring educational interventions for individuals with ASD, promising personalized support and enhanced learning experiences. Further real-world validation is necessary to optimize the effectiveness of this approach.**

***Index Terms*—Autism Spectrum Disorder, Machine Learning, ASD, toddlers, Special education, teaching methods, Autism**

I. Introduction

ASD is a neuro developmental condition with lifelong effects, and differing severity across a spectrum [1]. Data from 2020 suggests that roughly 1 in 36 children receive a diagnosis of ASD, marking a significant uptick of approximately 20 percent from 2018, when the diagnosis rate was reported as 1 in 44 [2]. Despite its common occurrence, the precise cause of autism remains unidentified, and there exists no definitive cure.

Timely detection is important in managing ASD, facilitating prompt initiation of interventions like behavioral therapies and educational approaches, which can greatly enhance outcomes [3]. Education for children with ASD remains a challenge, as they often require specialized support and resources [4]. Despite progress in research and diagnosis, the educational system continues to face challenges in meeting the needs of students with ASD, resulting in problems like school dropout rates and subpar academic performance.

Moreover, ASD showcases extensive variability, with each person displaying distinct characteristics. Consequently, educational strategies that prove effective for one individual may not be ideal for another. This poses a challenge in determining the most suitable teaching method for each child. The diversity inherent in ASD, as suggested by its name, necessitates a comprehensive approach to discerning the individual requirements of each child [5].

In the 1970s, evidence-based practice (EBP) emerged as a method to ensure that programs for children with autism are effective using scientific evidence. This approach has identified two main types of programs: comprehensive treatment models (CTMs) and focused intervention practices [6]. CTMs aim to enhance various aspects of a child's development, while focused intervention practices concentrate on specific skills or goals. By applying these EBPs, personalized education or intervention plans can be developed and consistently used in early intervention and school-based programs, leading to improved outcomes for students with autism [6].

Various intervention methods have been studied to meet the diverse needs of people with autism. Antecedent-based interventions, such as using visual cues, have been effective in reducing stereotypical behaviors like consistent repetitive, non-functional actions [7]. Technology-Aided Instruction and Intervention, specifically computer-assisted instruction, has been recognized as an EBP. However, not all technology-based interventions have been found to be effective [8].

Task Analysis has proven effective in training teachers for inquiry-based science instruction aimed at students with intellectual disabilities [9]. This highlights a direct link between training and teaching capability [9]. Pivotal Response Training (PRT), where play is used to help a child with their behavior, has shown effectiveness in enhancing complex social behaviors, language skills, and joint attention in children with autism. Moreover, these positive changes have been observed to persist over time [10]. Peer-mediated strategies, which leverage socially competent peers to model and reinforce appropriate behavior, have significantly improved social interactions by altering peer expectations and promoting peer effort [11]. The Picture Exchange Communication System (PECS) has been discovered to significantly improve social-communicative skills, particularly in adaptive behavior and

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unstructured environments. This makes it a valuable resource for non-verbal children with autism [12].

Through the utilization of machine learning algorithms, we can enhance the precision of identifying the educational requirements of children diagnosed with ASD. This, in turn, enables the implementation of more efficient and inclusive educational strategies. Various interventions are available for different subtypes of autism, with the effectiveness of these interventions depending on the specific characteristics of each child with ASD [13].

II. Related Work

Machine learning has demonstrated significant potential in addressing various aspects of ASD, including early detection, screening, and personalized educational interventions. Neural network approaches, such as those employed in the Modified Checklist for Autism in Toddlers-Revised (M-CHAT-R), made notable advancements, achieving high accuracy rates in early ASD detection among toddlers [14]. Furthermore, algorithms like logistic regression, when combined with feature selection techniques, have exhibited promise in ASD screening for adolescents and adults [15].

Autism screening typically involves administering a questionnaire designed to identify behavioral traits and certain individual characteristics, which are then used to generate a score indicative of the likelihood of autism [16]. Recent studies have simplified and enhanced ASD screening in toddlers using a neural network on data from nearly 15,000 toddlers. Researchers achieved 99.72% accuracy with just 18 questions from the M-CHAT-R screening tool, consistent across various demographics. This method could streamline ASD screening, reduce follow-up procedures and human error, and lead to earlier diagnosis and intervention [14].

Efforts to support children with Pervasive Developmental Disorders (PDD) have spurred the development of sophisticated systems like the Educational Activity Finder (EAF) and the Agent-based Educational Activity Discovery System [17] [18]. Both systems leverage Semantic Web technology and intelligent agents to tailor educational methods and materials to the unique needs of children with PDD. By utilizing semantic descriptions and ontology-based knowledge representation, these systems aim to comprehensively support parents, educators, and specialists in the care and education of these children.

Additionally, Duygu Ertugrul and Ali Ulusoy have introduced a knowledge-based medical expert system for pediatric patients, including those with PDD [19]. By leveraging ontology-based knowledge representation and semantic web technologies, this system enables remote consultation and monitoring of pediatric patients, especially in areas distant from medical service centers. Real-world case studies demonstrate the feasibility and effectiveness of employing intelligent systems to enhance pediatric healthcare delivery and support.

Dr. Fadi Fayez Thabtah's work in curating datasets containing responses, scores, and suitable interventions for various types of autism has laid a solid foundation for AI-driven intervention planning [20]. Furthermore, the integration of machine learning in identifying tailored teaching methods based on individual characteristics like behavior and communication has shown promise in improving learning experiences for autistic children [5].

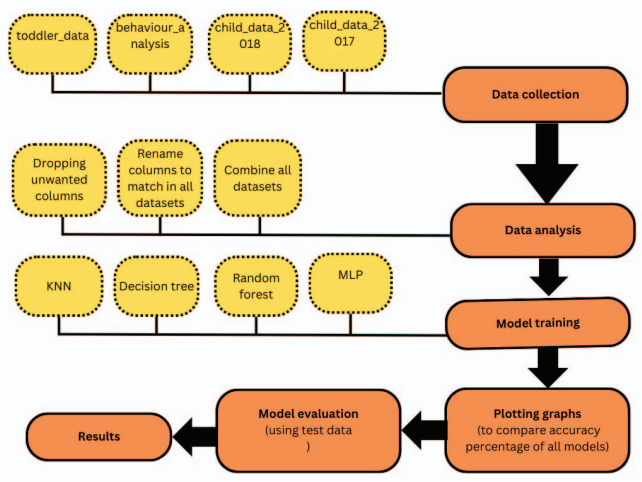


Fig. 1: Experimental Procedure Pipeline

In this study, we propose using Machine Learning's unique capabilities to create an AI capable of identifying the proper teaching methods for a child with ASD. There are 7 distinct educational methods that we will take into consideration: Technology-aided instruction, Antecedent-based intervention, PRT, Peer-mediated instruction, PECS, Task Analysis, and general education. Our study showcases Machine Learning's potential in tailoring educational interventions for individuals with ASD, promising personalized support, and enhancing their learning experiences. Further real-world validation of this approach is needed to optimize its effectiveness.

This AI will use extensive datasets to train and test in order to become an expert analyst of the needs of an autistic individual. After evaluating datasets and making a uniform constructed dataset, we split it into a train and test set. We then employed four algorithms (KNN, Decision Tree, Random Forest and MLP) to find the most efficient one. There were two algorithms which worked best (the MLP and Decision Tree algorithms), and which are most suitable for deducing teaching methods for children with ASD.

III. Materials and Methods

In this section, we describe the dataset used and the methodology utilized for conducting the experiments.

A. Dataset

Four datasets which can be referred from "https://github.com/SavarToteja/Autism-in-Toddlers" were combined for this study. The first dataset, "Autistic Spectrum Disorder Screening Data for Toddlers," was documented by Dr. Fadi Fayez Thabtah, Principal Lecturer in the Digital

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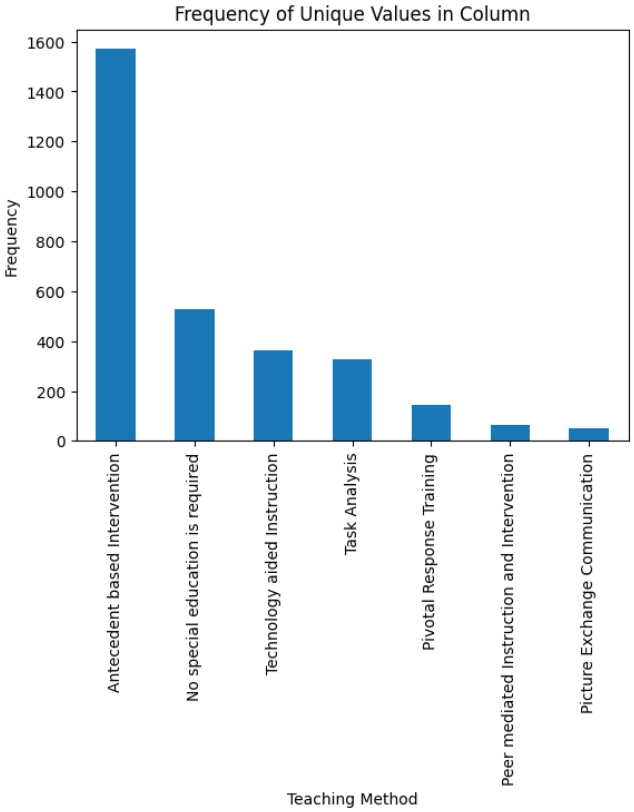


Fig. 2: Distribution of the Teaching Method column in the dataset

Technology department at Manukau Institute of Technology, New Zealand. The second dataset was "Behavior Analysis of Autism." Additionally, two versions of the "Autism Screening Child" dataset were used. The survey focused on toddlers under 3 years of age, where parents answered 10 yes-or-no questions regarding their child's behavior, forming 10 features named A1-A10. The remaining seven features were collected from the ASDTests screening app. Due to the insufficiency and general nature of individual datasets concerning clinical or screening autism spectrum disorder, all datasets were combined into a single uniform dataset. The final dataset contained 3043 samples with 17 features.

A target column was re-named "Preferred Education," across all datasets. This column contains seven categories: six for special teaching methods and one for no special education required. The sample distribution across various classes is shown in Figure 2. The category PECS has the smallest representation within the dataset, whereas "Antecedent Based Intervention" is the most prevalent category. The selection of these six teaching methods is based on specific binary conditions related to the autism traits data types A1-A10, as shown in figure 3 [21].

In preparation for analysis, the missing values of the dataset were imputed and the features were normalized. Furthermore,

Technology Aided Instruction	Antecedent based Intervention	Pivotal Response Training	Peer-mediated Instruction and Intervention	Picture Exchange Communication	Task Analysis	No special education is required
A5 =1, A9 =1 and A10 =0	A6 =1	A1 =1 and A8 =1	A5= 1, A4= 1 and A3 =1	A2 =1 and A9 =1	A7 =1	In any other condition

Fig. 3: Teaching method (Target) evaluation conditions

the dataset was partitioned into disjoint training and testing subsets, using the 80-20 ratio. No adjustments were made for data skewness prior to conducting the experiments.

B. Methodology

Seven different teaching methods were selected which provide a spread of appropriate educational channels. Firstly, Technology-aided instruction utilizes technology to allow children to learn at their own pace. Antecedent based intervention modifies the environment to eliminate interfering behaviors. PRT enhances motivation and communication skills. Peer-mediated instruction involves a peer without disabilities in teaching and social skill development. PECS encourages communication through visual symbols. Task Analysis breaks down tasks into smaller steps for easier comprehension. And finally, when no special education is required, the child does not need treatment for autism.

In the training datasets prior to machine learning, the children had already been assigned to a specific teaching method. This selection was made based on what weaknesses the student had, and which teaching method would help with those weaknesses the most. For example, a child that struggled to complete tasks that require multiple steps and appear complex would be assigned to Task Analysis training, where those steps are broken down into more understandable objectives.

In our study, for model training we employed four different types of algorithms. K-Nearest Neighbors (KNN) was chosen for its simplicity and effectiveness in instance-based learning. Decision Trees were chosen for their ease of interpretation and as a good baseline for tree based algorithms. We choose Random Forest algorithm for its hierarchical decision-making combined with ensemble learning that typically makes it a robust choice for many applications. We also wanted to explore an architecture that can capture non-linear relationships and our choice for this was the multi-layer perceptron (MLP). Each algorithm was chosen to assess model performance across a spectrum of data characteristics and learning tasks.

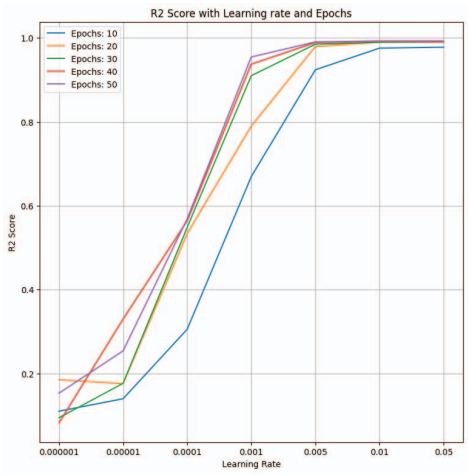
Training was done with different hyper-parameters for each of the algorithm. K-Nearest Neighbors (KNN) was trained with k values ranging from 1 to 20. The Decision Tree model was trained with max depth values from 1 to 7. The Random Forest model was tuned with two hyper-parameters: max depth in the range 1 to 7 and the number of trees in the range 10-100. The Multi-Layer Perceptron (MLP) algorithm was tuned with two hyper-parameter values: epochs with values

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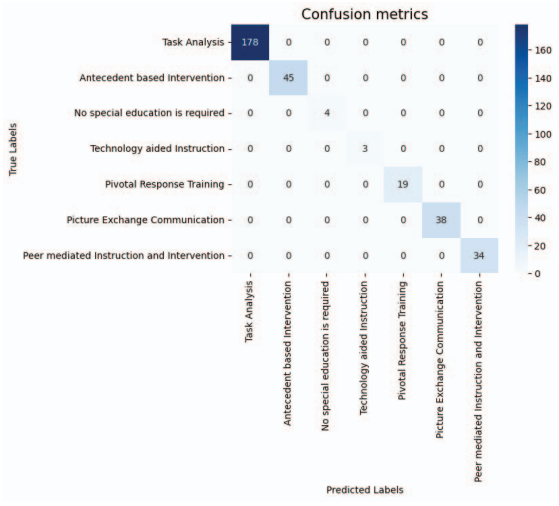
10-50 and learning rates between 0.05 and 0.000001. The performance of each algorithm with different hyper-parameters was evaluated on the validation dataset. Each model with best performing hyper-parameters were evaluated on the test dataset and the corresponding confusion matrix was used to visualize the performance of each model.

IV. Results

The results of the study revealed notable performance differences among the machine learning algorithms evaluated. During the training phase, the Multilayer Perceptron (MLP) algorithm achieved the highest accuracy of 99.3% on the validation set for the hyper-parameters learning rate 0.05 and 30 epochs as can be seen from Figure 4a.



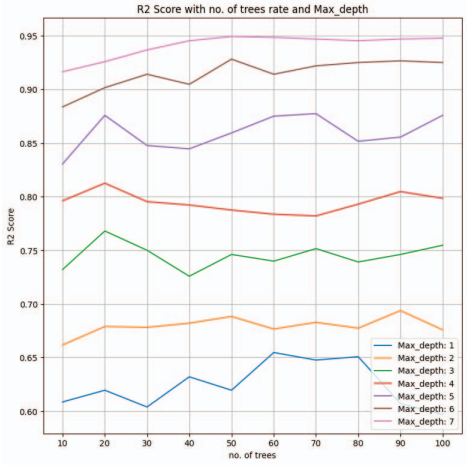
(a) Hyper-parameter tuning results for different values of learning rate and epochs



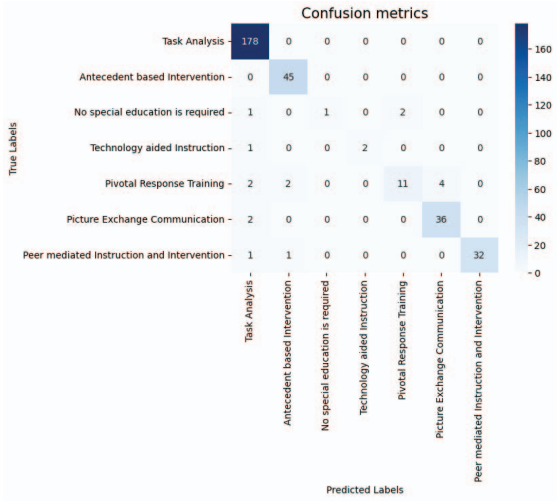
(b) Confusion matrix evaluated on test data

Fig. 4: MLP graph and confusion matrix

Following closely behind was the Random Forest algorithm, which attained a validation accuracy of 94.9% for the hyper-parameters of 50 estimators and a depth of 7 as shown in Figure 5a. The Decision Tree algorithm achieved an accuracy score of 94.4% for a depth of 7. In contrast, the K-Nearest Neighbors (KNN) algorithm had the highest validation training of 80.6% for a K value of 9 as shown in Figure 7a.



(a) Hyper-parameter tuning results for different values of max depth and no. of trees

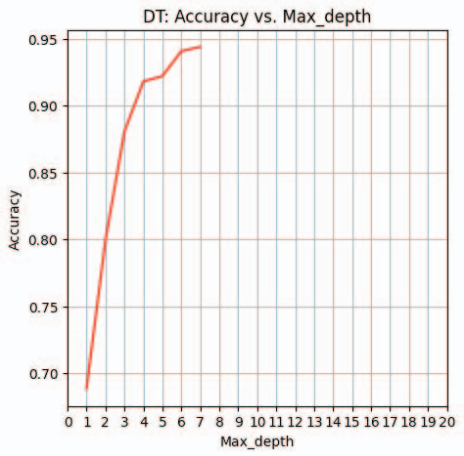


(b) Confusion matrix evaluated on test data

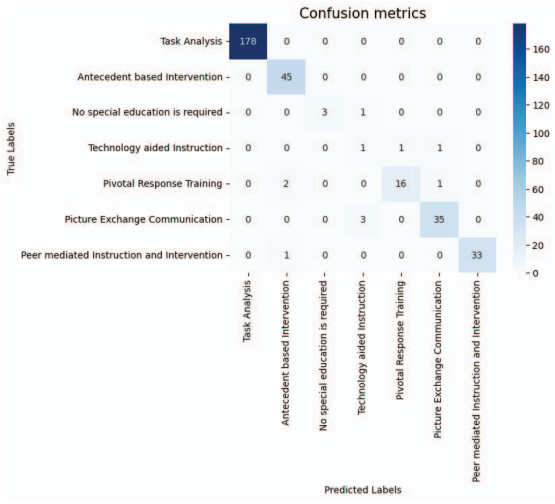
Fig. 5: Random forest graph and confusion matrix

Subsequently, the model underwent testing with a separate dataset to evaluate its real-world applicability. The results from the test set are shown in Figures 4b, 5b, 6b and 7b. The MLP algorithm maintained its high accuracy, achieving a perfect score of 100%. Similarly, the Decision Tree algorithm achieved an accuracy of 96.9%. The Random Forest algorithm demonstrated consistent performance during testing as well, with a test accuracy of 95%. Although the KNN algorithm showed a slight improvement in accuracy during testing, reaching 81.6%, it still fell short compared to the accuracy levels achieved by other algorithms.

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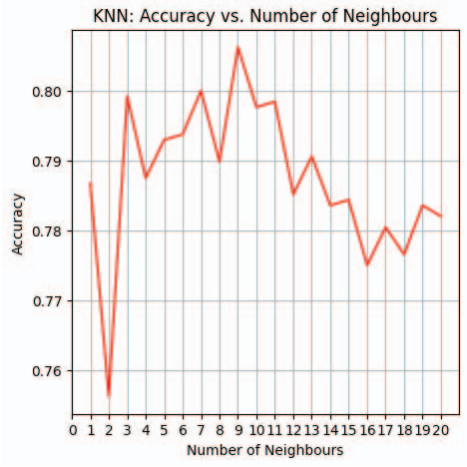


(a) Hyper-parameter tuning results for different values of max depth

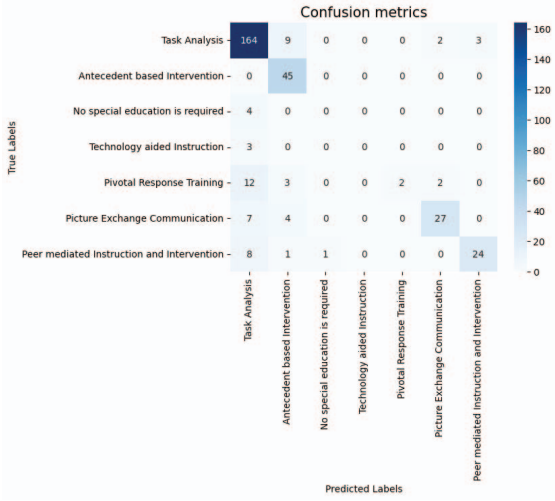


(b) Confusion matrix evaluated on test data

Fig. 6: Decision tree graph and confusion matrix



(a) Hyper-parameter tuning results for different K values



(b) Confusion matrix evaluated on test data

Fig. 7: KNN graph and confusion matrix

V. Discussion

In this section, we examine the implications of our findings for ASD intervention. The Multilayer Perceptron (MLP) algorithm achieved a perfect 100% accuracy in real-world testing, showing its potential for identifying optimal teaching methods for individuals with ASD.

MLP performance was evaluated across various learning rates and training epochs. As it can be seen in figure 4a, low learning rates (0.000001 to 0.0001) resulted in poor performance, particularly in early stages (10 to 30 epochs). For example, a learning rate of 0.000001 reached only 0.185 accuracy after 20 epochs, indicating underfitting. Higher learning rates improved performance significantly, with rates from 0.001 upwards achieving over 90% accuracy by 30 epochs. The highest accuracy (over 98%) was observed at learning rates of 0.01 and 0.05. Accuracy plateaued at higher epochs and learning rates, suggesting optimal performance is reached without further significant gains.

The Random Forest model's performance, based on varying tree depths and number of trees, is shown in Figure 5a. Shallow depths (e.g., depth 1) led to underfitting, with accuracies around 6%. Increasing tree depth improved accuracy, with depth 4 stabilizing above 78% and depth 7 peaking over 94% with 50 trees. Lower depths showed diminishing returns, indicating an optimal depth around 7.

Decision Tree model results also showed increased accuracy with greater depth, similar to Random Forest, but Random Forest achieved slightly higher accuracies (e.g., 94.875% at depth 7) due to its ensemble method reducing variance. This can easily be seen in Figure 6a

As shown in figure 7a, the k-nearest neighbors (KNN) model's performance varied with different k values, peaking at 80.625% accuracy with k=9.

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Testing the best models, MLP maintained 100% accuracy, Decision Tree achieved 96.9%, Random Forest 95%, and KNN 81.6%. These findings highlight the importance of careful model tuning to ensure effective real-world application.

VI. Conclusion

Our study evaluated the performance of automated techniques using machine learning in providing an assessment of which type of personalized education is optimal for children diagnosed with ASD. As ASD diagnoses continue to rise, the imperative to develop and implement effective, finely-tuned educational strategies becomes increasingly urgent. By incorporating Machine Learning (ML) algorithms into our research framework, we have not only demonstrated their potential but have also highlighted the practicality of identifying optimal teaching methodologies for the diverse ASD population.

Our research findings reveal the exceptional accuracy and effectiveness of ML algorithms, such as the Multi-Layer Perceptron (MLP), Decision Tree, and Random Forest, in predicting and recommending suitable educational interventions. These tools offer the promise of transforming the educational trajectories of children with ASD, providing customized support and substantially improving learning outcomes.

The real-world application of these strategies presents considerable challenges, including resource limitations and the need for comprehensive teacher training. Looking forward, continuous validation and refinement of our AI models through real-world applications and ongoing empirical research are essential. In anticipation of future advancements, conducting longitudinal studies to monitor the long-term effectiveness of these teaching methods is vital. Such research will allow for adjustments and improvements as the needs of children with ASD evolve. Integrating feedback from educators, parents, and caregivers into the ML models will ensure that the interventions remain relevant and continually adapt to real-world feedback.

Our study contributes to the evidence supporting the impact of automation using machine learning in enhancing educational outcomes and improving the quality of life for individuals with ASD. This helps with inclusive and effective educational practices, empowering children with ASD to achieve their full potential and ensuring a brighter future.

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